NETWORKING

TCP 3-Way Handshake Visual Guide

How two machines agree to talk reliably before a single byte of data flows.

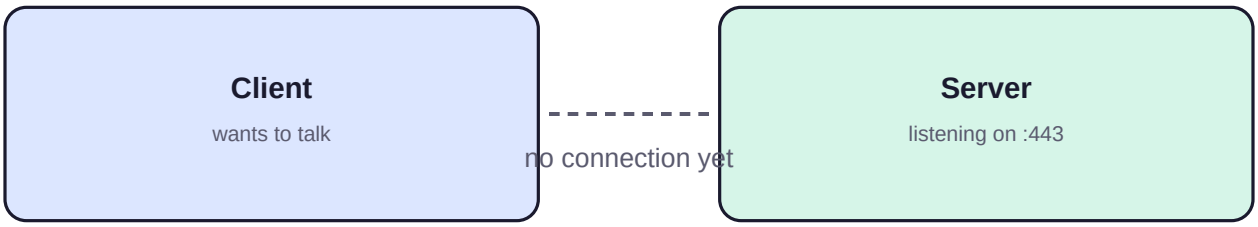
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Why a Handshake at All?

TCP is reliable: ordered, acknowledged, connection-oriented. That starts with agreement.

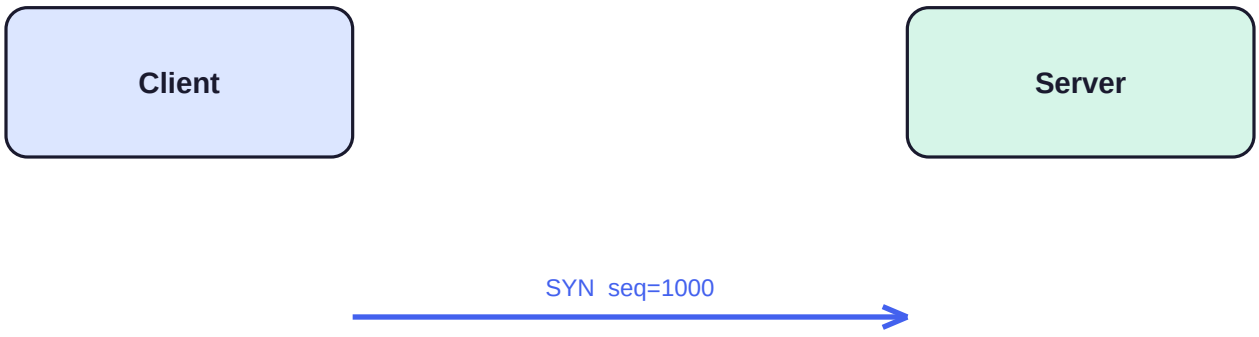


UDP just fires datagrams. TCP negotiates first — sequence numbers, windows, options — so nothing is lost or duplicated.

STEP 2/6

Step 1 — SYN: 'Can we talk?'

Client picks a random initial sequence number (ISN) and sends SYN.



SYN = synchronize sequence numbers.

Client says: 'I will number my bytes starting at 1000.'

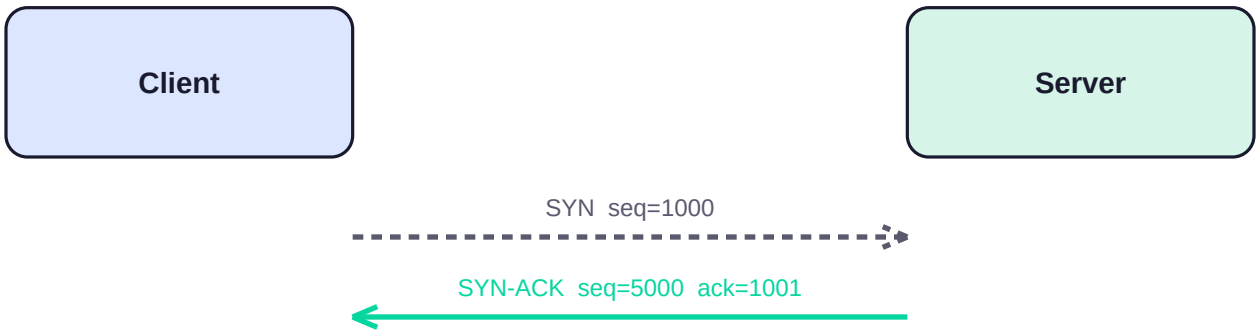
Packet: SYN

seq=1000 • SYN flag set • no data

STEP 3/6

Step 2 — SYN-ACK: 'Yes, and here's my number'

Server acknowledges the SYN and sends its own ISN.



ack=1001 means: 'I expect your next byte to be #1001.'

seq=5000 means: 'I will number MY bytes starting at 5000.'

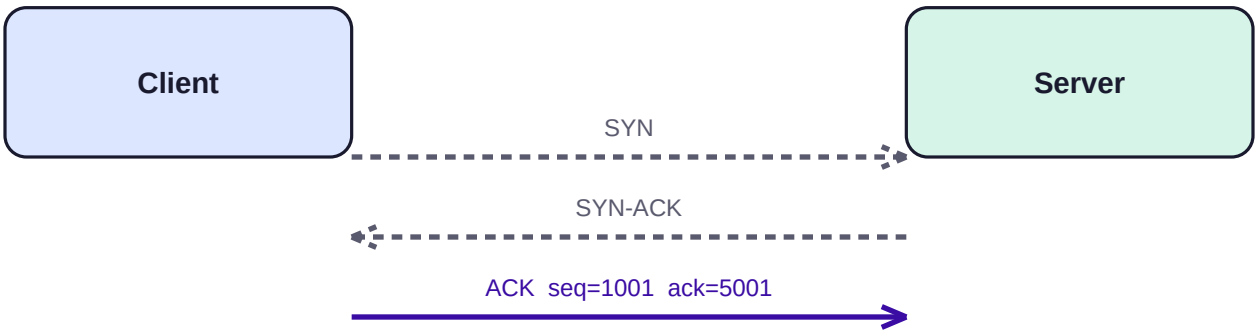
Packet: SYN-ACK

seq=5000 ack=1001 • SYN+ACK flags

STEP 4/6

Step 3 — ACK: 'Got it, we're connected'

Client acknowledges the server's SYN. Connection: ESTABLISHED.



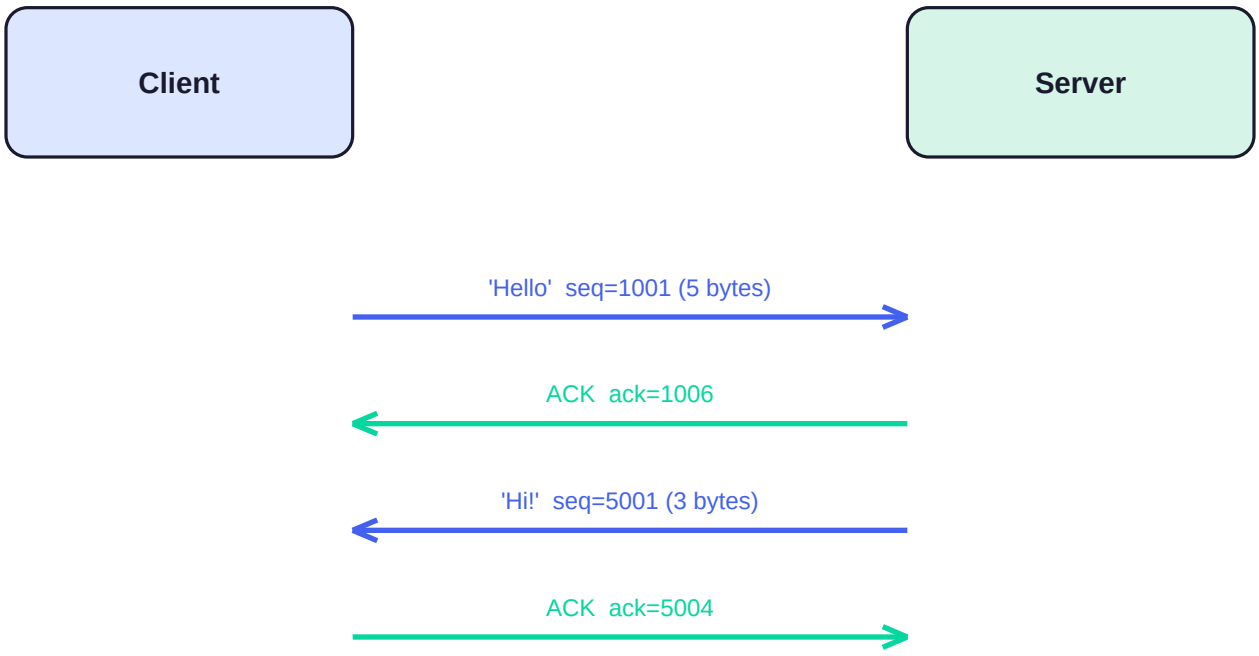
ESTABLISHED ✓

Both sides agreed on sequence numbers. Data can now flow.

STEP 5/6

After the Handshake: Reliable Data Flow

Every byte numbered, every segment acknowledged.



Lost a packet? The sender retransmits after a timeout.

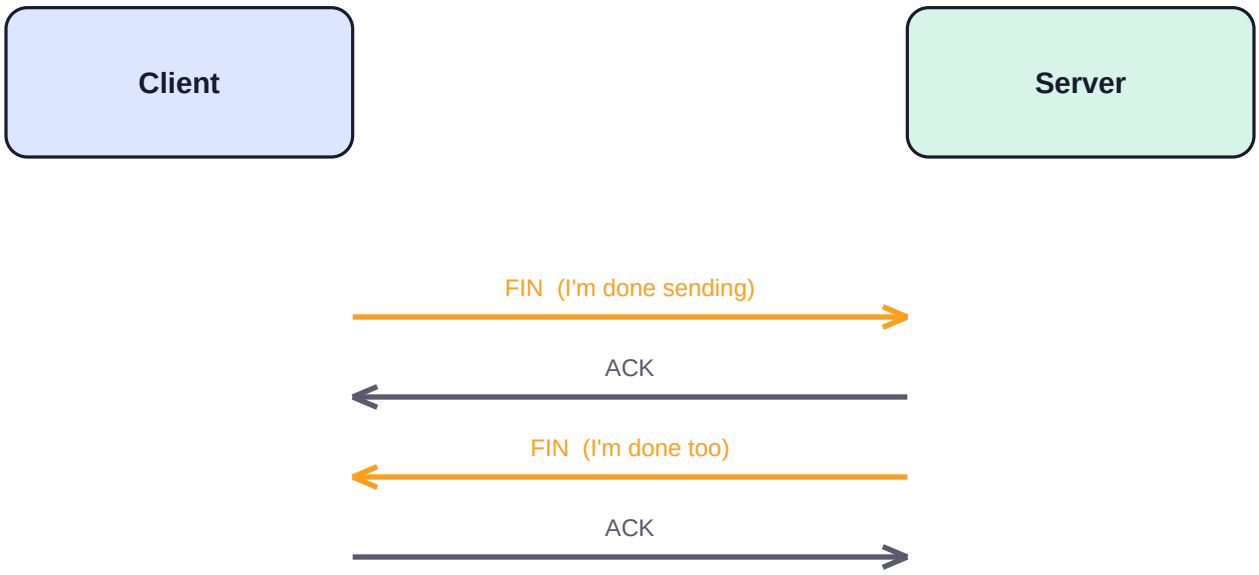
Arrived out of order? The receiver reorders by sequence number.

Receiver overwhelmed? The window size tells the sender to slow down (flow control).

STEP 6/6

Closing: The 4-Way Goodbye

Each side closes independently with FIN.



TIME_WAIT: the closer waits 2×MSL so stray packets die before the port is reused. Prevents old data corrupting new connections.

STEP ✓/✓

Key Takeaways

TCP Handshake — Remember

- SYN → SYN-ACK → ACK: both sides agree on initial sequence numbers before data flows.
- ack=N means 'I expect byte N next'; seq=N numbers the sender's bytes.
- Reliability = sequence numbers + acknowledgments + retransmission + flow control.
- Closing takes 4 steps (FIN/ACK each way); TIME_WAIT guards against stray packets.
- Cost: 1 round-trip before any data — the reason for TCP Fast Open and QUIC.

Keep exploring

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